

One Policy for the Whole Arm: Region-Conditioned Distillation for Safe Robotic Bed-Bathing

Authors: TBD

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Abstract

Robotic assistance with bed-bathing requires a manipulator to wipe the skin of a supine person completely, systematically, and with clinically safe contact forces. Reinforcement learning (RL) can produce competent per-region wiping policies, but deploying a separate network for every body part is impractical, and the standard contact-counting reward optimizes the wrong objective—it clears targets aggressively without regard to coverage or force safety. We make three contributions. First, the *Comfort reward*: a structured wiping reward that shapes motion along a self-supervised arm axis—extended here with a 2-D (length $\times$ circumference) coverage term built from the same SVD’s second principal axis—producing complete, force-safe sweeps. Second, a *single distilled policy conditioned on region and a one-bit comfort dial* that replaces eight per-region specialists, matches or exceeds them in every region, and beats a flat multi-task policy by **+33% targets cleared (positive in 12/12 region $\times$ seed pairs, exact sign test $p = 0.0005$)** while keeping contact force in the therapeutic band $2\text{--}2.7\times$ more often. Third, the dial bit exposes a *caregiver-selectable gentle $\leftrightarrow$ thorough* behavior: in a paired same-network test, the Comfort-reward (Gentle) setting raises therapeutic-band adherence by **+34% (10/12 pairs, $p = 0.039$)** at an 8% coverage cost. Across the full method ladder, learning dominates a scripted geometric controller on every wiping-quality metric—the scripted sweep traces near-perfect geometry yet fails sustained safe contact—and the reward design decides not how *much* the robot wipes but how *safely*.

1 Introduction and Problem Definition

A UR5 manipulator holds a soft tool and cleans a target region of a supine human arm in simulation (PyBullet). Each region carries up to 15 cleanable *targets*; a target is *cleared* when the tool passes within `clear_radius` = 0.022 m while in contact. Episodes last 150 steps. The observation is a 90-D vector (robot joint state, end-effector pose, tool-skin contact geometry, region/target encoding); the action is a 6-D continuous end-effector command.

Why raw clear-count is the wrong objective. Assistive bathing is a *coverage-and-safety* task, not a target-collection game. A caregiver wipes the whole region in order, maintain-

ing gentle, sustained skin contact. A reward that simply counts contacts encourages fast, aggressive dabbing that inflates the clear-count while leaving gaps and spiking force. We therefore evaluate on a battery of behavior metrics (Sec. 3.3) and design a reward—the *Comfort reward*—for the true objective.

Why one policy, not four. Region-specialist policies are easy to train but impractical to deploy: a robot would need to store and select among a library of networks. We ask whether a *single* context-conditioned policy, distilled from the specialists, can replace the whole library without loss—the paper’s central generalization question.

2 Method

2.1 Teacher policies

One PPO policy (RLlib 1.13) per (region, reward) pair. Policy/value net `fcnet_hiddens` = [100, 100], `tanh`; `train_batch_size` = 19200, `num_sgd_iter` = 50, `sgd_minibatch_size` = 128. Trained in $\sim 250\text{k}$ -step chunks with best-checkpoint archiving.

2.2 Reward design

We compare two reward designs on identical dynamics. The baseline is the naive contact-counter; the proposed *Comfort reward* keeps its base terms and adds coverage-and-safety shaping.

Baseline: Contact-count reward R_{cc} . The original reward shipped with the public bed-bathing codebase (git 96fb9af):

$$R_{cc} = w_d (-d_{\text{arm}}) + w_a (-\|a\|^2) + w_c c_t, \quad (1)$$

with $(w_d, w_a, w_c) = (1.0, 0.01, 5.0)$, where d_{arm} is tool-arm distance, a the action, and c_t the targets cleared at step t . It rewards *how many* targets are cleared but has no notion of *where* along the arm the tool is or whether coverage is systematic.

Proposed: Comfort reward R_{comf} . We keep the base terms and add four coverage-shaping terms:

$$R_{\text{comf}} = w_d r_{\text{dist}} + w_a r_{\text{act}} + w_c r_{\text{clr}} + w_s r_{\text{sweep}} + w_w r_{\text{win}} + w_e r_{\text{end}} + w_v r_{\text{cov}}, \quad (2)$$

with weights $(w_d, w_a, w_c, w_s, w_w, w_e, w_v) = (1.0, 0.01, 5.0, 1.0, 1.0, 1.0, 1.0)$, $r_{\text{act}} = -\|a\|^2$, and

Term	What it rewards	Weight
r_{dist}	on-surface proximity (two-phase)	1.0
r_{act}	$-\ a\ ^2$ (effort penalty)	0.01
r_{clr}	targets cleared this step c_t	5.0
r_{sweep}	monotone progress along arm axis	1.0
r_{win}	contact/clears near current s	1.0
r_{end}	evidence-gated end-of-pass bonus	1.0
r_{cov}	first visit to each (s, w) grid cell	1.0

Table 1: Terms of the Comfort reward R_{comf} (Eq. 2). The Contact-count baseline R_{cc} (Eq. 1) uses only the first three rows. The four shaping terms turn a target-collector into a systematic, force-safe wiper.

$r_{\text{clr}} = c_t$. Table 1 summarizes each term; the four shaping terms are detailed below.

Two-phase distance (r_{dist}). $r_{\text{dist}} = -d_{\text{arm}}$ while reaching, switching to the along-surface distance once wiping, so the policy is rewarded for staying on-surface rather than merely approaching.

Self-supervised sweep axis (r_{sweep}). Let u be the principal axis of the target cloud (SVD, refreshed every ~ 200 steps) and p the tool position. The normalized sweep coordinate is $s = \text{clip}((\langle p - \bar{p}, u \rangle - s_{\text{min}})/\text{span}, 0, 1)$. With $\Delta s_{\text{dir}} = \rho(s_t - s_{t-1})$ for pass direction $\rho \in \{\pm 1\}$,

$$r_{\text{sweep}} = \begin{cases} k_f \min(\Delta s_{\text{dir}}, \delta), & \Delta s_{\text{dir}} > \varepsilon \\ -k_b \min(-\Delta s_{\text{dir}}, \delta), & \Delta s_{\text{dir}} < -\varepsilon \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

active only while in contact ($k_f=0.5, k_b=0.75, \varepsilon=0.01, \delta=0.05$). It rewards monotone progress along the arm and mildly penalizes backtracking.

Local window (r_{win}). $r_{\text{win}} = 0.5 n_{\text{contact}}^{\text{win}} + 0.25 n_{\text{clr}}^{\text{win}}$, crediting contact/clears near the current s (discourages teleporting between distant targets).

Evidence-gated end-of-pass bonus (r_{end}). $r_{\text{end}} = B$ ($B=4.0$) when a pass reaches an end ($s \geq 0.90$ forward or $s \leq 0.10$ backward) and the pass accumulated real coverage evidence (≥ 1 in-window hit/clear); otherwise 0. The gate prevents farming the bonus by sweeping empty space. A soft-contact latch with hysteresis (6 steps) stabilizes the wipe/exit phase against brief force dropouts.

2-D (s, w) coverage (r_{cov}). For this elongated target geometry the *second* principal axis of the same SVD is the circumferential (“column”) direction; projecting the tool onto it yields a normalized width coordinate $w \in [0, 1]$. The first contact or clear in each unvisited cell of a 4×4 (s, w) grid earns $r_{\text{cov}} = 1$ once per episode ($w_v=1.0$ selected by grid search over $\{0, 0.5, 1, 2\}$: 6.08 vs. 5.68 targets cleared at $w_v=0$). The 1-D sweep terms reward progress *along* the arm only; this term is what gives the policy an incentive to cover *across* it.

2.3 Region-conditioned student

A single Gaussian MLP policy conditioned on the region:

input = [obs(90), context]
context = region one-hot(4) [+ dial bit(1)]
backbone = MLP(128, 128), tanh
heads = mean(6); log-std(6) $\in [-5, 2]$

a diagonal Gaussian $\pi_{\theta}(a | o, c) = \mathcal{N}(\mu_{\theta}, \text{diag } \sigma_{\theta}^2)$. At deployment the one-hot selects the current body part and the optional *comfort dial* bit $\kappa \in \{0, 1\}$ selects behavior; one network covers the whole arm.

2.4 Distillation

Each teacher π_r is rolled out in its own region to record $(o, \mu_r(o), \log \sigma_r(o), c(r))$, i.e. the observation, the teacher’s diagonal-Gaussian action mean and log-std, and its context vector. Pooling all regions gives $\mathcal{D} = \bigcup_r \mathcal{D}_r$. The student π_{θ} is fit by supervised distillation—a behavior-cloning term on the mean plus a distribution-matching term on the full Gaussian:

$$L(\theta) = \underbrace{\|\mu_{\theta} - \mu_T\|^2}_{\text{behavior clone}} + \beta \underbrace{\text{KL}(\mathcal{N}_T \| \mathcal{N}_{\theta})}_{\text{uncertainty match}}, \quad (4)$$

where, for diagonal Gaussians with per-dimension means μ and log-stds $\ell = \log \sigma$,

$$\text{KL}(\mathcal{N}_T \| \mathcal{N}_{\theta}) = \sum_{i=1}^6 \left[\ell_{\theta,i} - \ell_{T,i} - \frac{1}{2} + \frac{\sigma_{T,i}^2 + (\mu_{T,i} - \mu_{\theta,i})^2}{2 \sigma_{\theta,i}^2} \right]. \quad (5)$$

We minimize the batch mean of Eq. 4 with Adam (lr 10^{-3} , $\beta=1$), matching both the teacher’s action mean and its uncertainty. We compare four distillation sources: **Comfort** (4 Comfort-reward teachers), **Contact-count** (4 contact-counting teachers), **dial-conditioned** (all 8 teachers + a comfort-dial bit that switches behavior on demand), and **flat** (all 8 pooled, no bit—an irreversible blend).

Algorithm 1: Region-conditioned distillation

Input: teachers $\{\pi_r\}_{r \in R}$; context map $c(r)$; rollout length N ; weight β ; (optional) dial values $\{\kappa_r\}$.
1. **Collect:** for each region r , roll out π_r for N steps; record $(o, \mu_r(o), \log \sigma_r(o), c(r))$ where $\mu_r, \log \sigma_r$ are the teacher’s Diag-Gaussian parameters. Append dial bit κ_r to $c(r)$ if given.
2. **Pool:** $\mathcal{D} = \bigcup_r \mathcal{D}_r$.
3. **Train** $\pi_{\theta}(a | o, c)$ by minibatch Adam (10^{-3}) on the mean of Eq. 4 (BC + β KL, Eq. 5).
Output: single conditioned student π_{θ} .

3 Experimental Design

Regions (4): forearm back/front, upper-arm back/front.

3.1 Methods and baselines

We evaluate three reward classes—Comfort RL, Contact-count RL, and a non-RL scripted geometric sweep—each as *both* a per-region specialist *and* a distilled student, plus:

- **Dial student (proposed):** *one* policy distilled from all eight RL specialists, conditioned on region and a one-bit comfort dial; evaluated at both dial ends, *Gentle* ($\kappa=0$, Comfort behavior) and *Thorough* ($\kappa=1$, Contact-count behavior).
- **Comfort / Contact-count specialists / students:** per-region PPO teachers and their single-family region-conditioned distilled students.
- **Scripted specialist / student:** the geometric sweep, and a student behavior-cloned from it (does a non-RL controller generalize when cloned?).
- **Flat baseline:** a *single* PPO policy trained jointly on all four regions with the Comfort reward and *no* region conditioning—the direct multi-task alternative to distillation.

3.2 Evaluation protocol

Every method is scored by identical reducers (`metrics.report.episode_metrics`), 200 episodes per region (100 for the deterministic scripted controller), force band $[2, 12]N$, `clear_radius = 0.022m`, and **3 training seeds** for every learned method. For significance we use exact sign tests over the 12 region $\times$ seed pairs plus a conservative per-seed paired *t*-test—the seed, not the episode, is the statistical unit (episode-level pooling would be pseudoreplication). Every number below is emitted by `analyze_final.py` from the raw metrics CSV.

3.3 Evaluation metrics

Per step t an episode yields normal force f_t , an in-contact flag, sweep coordinate $s_t \in [0, 1]$, per-step clears k_t , and action a_t ; $C = \{t : \text{contact}\}$ denotes the in-contact steps and T the episode length. Table 2 defines the curated headline set with exact formulas.

4 Results

4.1 Headline: one network with a caregiver-selectable gentle $\leftrightarrow$ thorough dial

The dial student—a single network conditioned on region and one comfort-dial bit—spans the coverage $\leftrightarrow$ gentleness trade-off at deployment time. Averaged over 4 regions $\times$ 3 seeds: **Thorough mode 6.80 cleared / 0.156 in-band, Gentle mode 6.17 cleared / 0.198 in-band**, with the ordering consistent on all four regions (Table 4). At each end the dial matches or exceeds the corresponding single-reward student—it is not a

compromise but both behaviors in one deployable policy. Table 3 reports every method across all four arm regions.

4.2 The distilled policy beats the flat multi-task baseline

The dial in Thorough mode clears **6.80** vs **5.11** for the flat baseline—**+33%**, positive in **12/12 region $\times$ seed pairs** (exact sign test $p = 0.0005$), consistent per seed (6.43/6.81/7.14 vs 5.26/5.37/4.69). The Comfort-distilled student alone gains +1.16 clears/seed (per-seed paired $t = 3.76$, $n = 3$; 10/12 pairs, sign $p = 0.039$). Conditioning also buys *safety*: every conditioned student keeps force in-band $2\text{--}2.7\times$ more than the flat baseline (0.153–0.198 vs 0.074). One network replaces four specialists and beats the unconditioned alternative on both axes.

4.3 Distillation $\geq$ specialists—and rescues weak teachers

Students match their specialists where those are strong and repair them where they are weak (cleared/15, 3-seed means): Comfort teacher $\rightarrow$ student 4.99 $\rightarrow$ **6.23** on forearm front and 5.19 $\rightarrow$ **5.96** on upper-arm front; Contact-count 4.95 $\rightarrow$ **5.69** on upper-arm front. No region regresses by more than 0.3. Grand means: students 6.17–6.80 vs specialists 5.79–6.38—pooled-data distillation regularizes.

4.4 The reward style determines gentleness: a paired same-network test

Because the dial student is *one* network, flipping the bit is a perfectly paired comparison—same weights, seeds, and regions; only the dial input differs. Switching to the Gentle (Comfort) setting raises **in-band force fraction by +34%** (better in 10/12 region $\times$ seed pairs, exact sign test $p = 0.039$) at an 8% coverage cost (3/12, n.s.). On the composite *safe clears* (cleared $\times$ in-band) the Comfort-distilled student leads by +28% (8/12 pairs, suggestive). Raw clear-count ordering is reversed (the Thorough/Contact-count setting clears more): the reward design does not make wiping more *productive*, it makes it *safer*—and the dial exposes exactly that trade-off to the caregiver.

4.5 Learning and conditioning both matter: the method ladder

Grand means over 4 regions $\times$ 3 seeds form a monotone ladder: non-RL scripted **4.56** cleared / **0.039** in-band $<$ flat RL **5.11** / 0.074 $<$ specialists 5.79–6.38 / 0.15–0.17 $<$ distilled students **6.17–6.80** / 0.15–0.20. The scripted controller is not merely less productive: it fails the wiping-*quality* battery—in contact only 25–41% of steps (vs 51–74% for the dial), in-band force collapsing to 0.020/0.002 on the upper arm—while

Metric	Definition	What it captures
Targets cleared	$\sum_t k_t$ (distinct, out of 15)	task performance
In-band force fraction	$ \{t \in \mathcal{C} : f_t \in [2, 12] \text{ N}\} / \mathcal{C} $	contact-force safety
Contact-time fraction	$ \mathcal{C} / T$	sustained skin engagement
Sweep consistency	(forward-progress wipe steps) / (wipe steps)	systematic, monotone motion
Clear s -coverage	$ \{\text{arm-axis bins with a clear}\} / B, B=10$	spatial spread along the arm
Clear t -coverage	$ \{\text{time bins with a clear}\} / B, B=10$	temporal spread (not bursty)
Second-half fraction	$(\sum_{t>T/2} k_t) / \sum_t k_t$	sustained, not front-loaded
Clear spatial order	fraction of consecutive clears moving in the dominant s -direction	orderly sweep (1.0) vs. random (0.5)
Mean stroke length	mean length of maximal in-contact, constant-direction sliding runs	wipe (long) vs. poke (short)
Contact bouts	number of separate contact episodes	poking fragments contact (higher = worse)
Slide fraction	$ \{t \in \mathcal{C} : \Delta s_t > 0.008\} / \mathcal{C} $	sliding vs. tapping in place
Peak force	$\max_{t \in \mathcal{C}} f_t$	worst-case contact force (lower is safer)

Table 2: Evaluation metrics with exact definitions. Two families are deliberately *excluded* from cross-method comparison. *Pass completions* counts firings of the Comfort reward’s own end-of-pass gate, so it measures one method’s internal machinery. *Mean/p95 force reward not touching*—the scripted controller has the lowest mean force precisely because it grazes—so force safety is reported as in-band fraction and peak force only. Remaining metrics (task success, action effort/smoothness, stall fraction) are retained in the appendix.

Method	cleared / 15 (in-band force fraction), 3-seed mean				
	forearm back	forearm front	upper-arm back	upper-arm front	mean
Comfort specialist	6.25 (0.150)	4.99 (0.218)	6.75 (0.164)	5.19 (0.159)	5.80 (0.173)
Contact-count specialist	6.72 (0.121)	7.05 (0.256)	6.80 (0.112)	4.95 (0.128)	6.38 (0.154)
Student $\leftarrow$ Comfort	6.15 (0.154)	6.23 (0.282)	6.74 (0.172)	5.96 (0.162)	6.27 (0.193)
Student $\leftarrow$ Contact-count	6.42 (0.117)	7.77 (0.256)	6.82 (0.110)	5.69 (0.128)	6.68 (0.153)
Dial @Gentle	6.35 (0.168)	6.15 (0.311)	6.62 (0.165)	5.55 (0.146)	6.17 (0.198)
Dial @Thorough	6.95 (0.121)	7.65 (0.278)	7.02 (0.106)	5.56 (0.120)	6.80 (0.156)
Flat (multi-task)	5.00 (0.073)	5.20 (0.059)	6.43 (0.109)	3.80 (0.058)	5.11 (0.075)
Scripted (non-RL)	6.27 (0.087)	4.48 (0.050)	4.28 (0.020)	3.20 (0.002)	4.56 (0.040)

Table 3: **Main comparison** across all four arm regions (3 seeds each). Each cell is targets cleared out of 15 with the therapeutic-band (2–12 N) force fraction in parentheses; bold marks the best coverage (per region) and the best in-band fraction. The conditioned students and the comfort dial dominate the flat multi-task and non-RL scripted baselines on both coverage and in-band force; the dial’s two ends bracket the coverage $\leftrightarrow$ gentleness trade-off. Numeric companion to Fig. 1.

Region	Gentle clr / in-band	Thorough clr / in-band
forearm back	6.35 / 0.168	6.95 / 0.121
forearm front	6.15 / 0.311	7.65 / 0.278
upper-arm back	6.62 / 0.165	7.02 / 0.106
upper-arm front	5.55 / 0.146	5.56 / 0.120

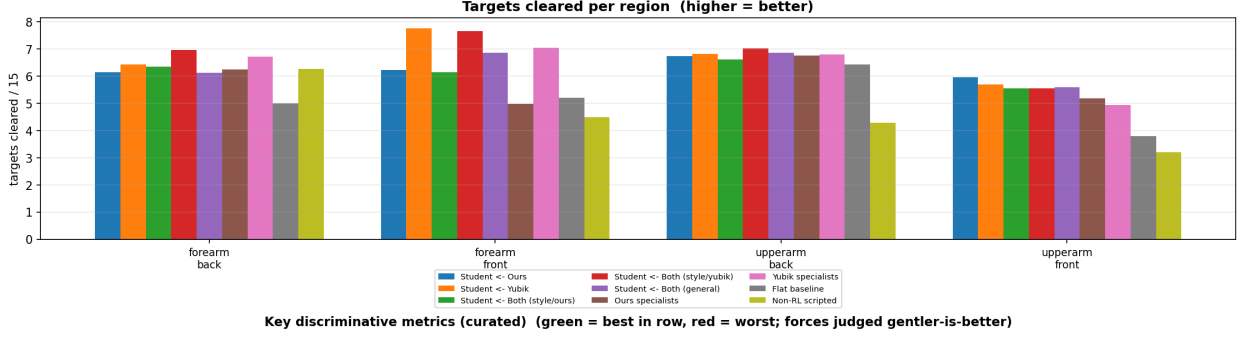
Table 4: The comfort dial per region (3-seed means): the Gentle end is gentler (higher in-band) in 4/4 regions, the Thorough end clears $\geq$ in 4/4.

its near-perfect clear spatial order (0.997) shows it traces textbook geometry. Raw clear-count and geometric orderliness are mirages of wiping; sustained safe contact had to be learned. Crucially, **distillation cannot repair it**: cloning the scripted controller into a student adds a little coverage (4.56 $\rightarrow$ 5.05 grand mean) but leaves in-band force at ~ 0.04 , versus ~ 0.19 for the RL students. Distillation *amplifies* what a teacher already does; it cannot inject safe contact a teacher never had—so the generalization is a property of the learned behavior, not of the distillation mechanism.

5 Limitations

Reachability. One angular column of each *back* region faces the bed—the supine limb’s support surface. Per-target IK over a dense orientation grid, dynamic clearability under three independent controllers, robot base-pose sweeps, and arm-posture sweeps all show no configuration of this environment can wipe it (the humanoid lacks a forearm pronation DOF). This caps targets-cleared near 10/15 on back regions and keeps `task.success` (69% threshold) near zero for *all* methods; cleaning the support-facing surface requires physically re-positioning the limb—a caregiver action our deployment pipeline assigns to its classical phases. **Force spikes.** Peak contact forces reach 47–68 N for all methods; bounding worst-case force (force-rate shaping or a controller-level clamp) is required before hardware trials. **Statistics.** Sign tests over 12 region $\times$ seed pairs are exact, but per-seed $n = 3$; per-method gaps quoted without a p -value are directional. **Scope.** Results are simulation-only (rigid humanoid, quasi-static scene); sim-to-real transfer is future work.

Assistive bed-bathing: one distilled student vs. specialist teachers (Ours / Yubik) and baselines



	Student <- Ours	Student <- Yubik	Student <- Both (style/ours)	Student <- Both (style/yubik)	Student <- Both (general)	Ours specialists	Yubik specialists	Flat baseline	Non-RL scripted
Targets cleared /15	6.270	6.675	6.166	6.796	6.358	5.793	6.382	5.106	4.558
p95 force (N)	4.162	4.258	4.303	4.004	4.339	4.419	4.329	3.163	2.133
In-band force frac	0.193	0.153	0.197	0.156	0.193	0.173	0.154	0.074	0.039
Pass completions	0.654	0.725	0.652	0.730	0.614	0.657	0.655	0.473	0.115
Sweep consistency	0.184	0.201	0.172	0.206	0.172	0.184	0.195	0.181	0.046
Drift rate	0.078	0.099	0.088	0.098	0.084	0.000	0.000	0.000	0.138
Clear s-coverage	0.342	0.351	0.343	0.355	0.341	0.336	0.354	0.286	0.260
Clear t-coverage	0.343	0.340	0.342	0.338	0.341	0.336	0.332	0.282	0.220
2nd-half frac	0.110	0.099	0.107	0.087	0.110	0.105	0.079	0.053	0.003

Figure 1: Method comparison across four arm regions and three seeds. Top: targets cleared per region. Bottom: curated discriminative metrics (green = best in row, red = worst).

6 Conclusion

A single distilled policy, conditioned on region and a one-bit comfort dial, wipes all four arm regions, beats the flat multi-task baseline in every region×seed pair (+33%, $p = 0.0005$), and exposes a caregiver-selectable gentle↔thorough behavior whose Gentle end raises therapeutic-band adherence by 34% in a paired same-network test ($p = 0.039$). Across the method ladder, learning dominates scripting on every wiping-quality metric, and conditioned distillation dominates flat multi-task RL on both coverage and safety. The Comfort reward does not decide how *much* a robot wipes—it decides how *safely*; distillation turns that choice into a runtime-switchable property of one deployable network.